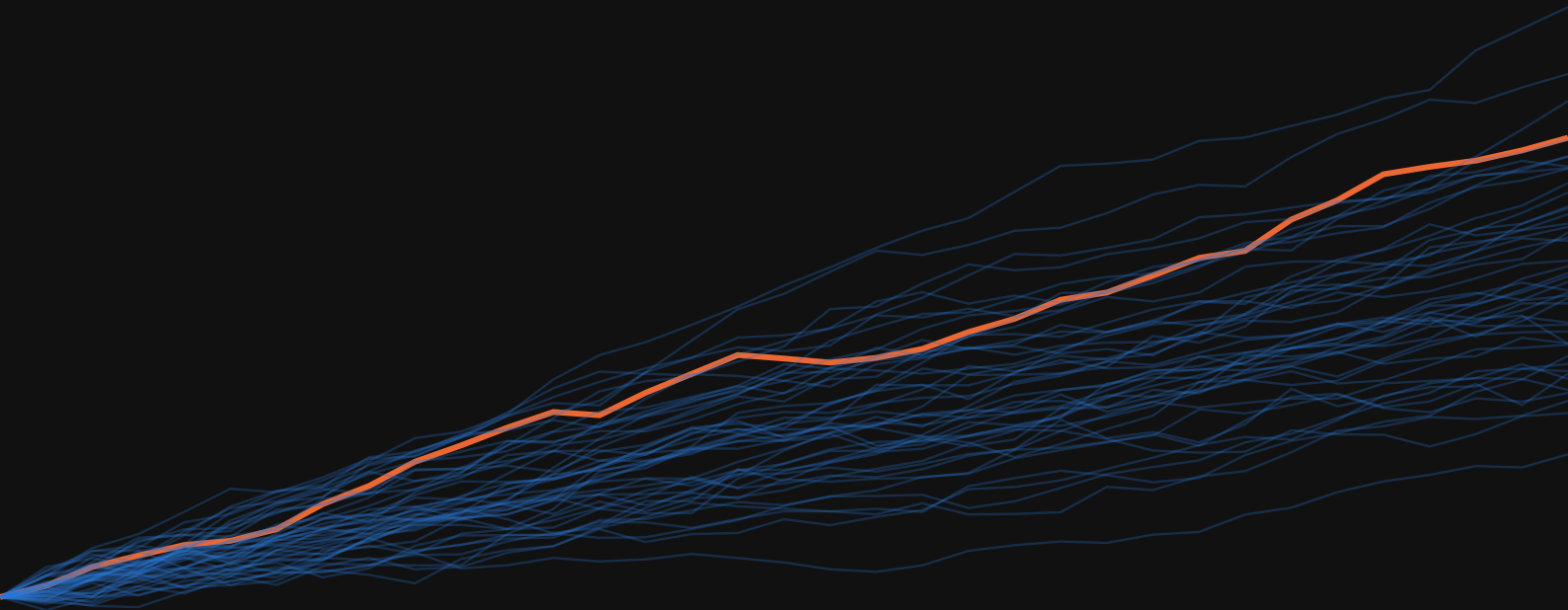


The Champion Problem

Can Artificial Intelligence Predict
Who Wins the League?

Predictive analytics, luck and the human factor in football



46 simulated title races • one champion

Dennis Buchbauer

Predictive & Betting Analytics

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ABSTRACT

Can a machine tell us in August who lifts the trophy in May?

Every summer, supercomputers publish their title predictions, and every summer the headlines treat them as forecasts of the future. In 2024/25, Opta's model gave Manchester City an 82% chance of winning the Premier League. Liverpool, at 5%, won it. In the same season, the pre-season favourites of La Liga and Serie A – Real Madrid at 86% and Inter at 79% – also failed. One year later, Arsenal, given 0.13% three seasons earlier, became champions.

This paper asks a simple question: **is it possible today to determine the league champion with the power of AI, or is the human factor still too strong?** I combine three sources of evidence: a scorecard of 15 recent pre-season model predictions in Europe's top leagues and five major tournaments; a review of the academic literature on luck and skill in football; and an original Monte Carlo simulation of 20,000 seasons per scenario that separates the uncertainty a model can remove from the uncertainty it cannot.

The answer is clear. **AI can identify the most likely champion, but it cannot determine the champion.** Football is the most upset-prone of the major team sports, and almost half of all goals involve an element of chance. My simulation shows that even a model with perfect knowledge of every team's true strength would name the wrong champion in roughly one of three tight title races. The human factor – injuries, coaching changes, crowds, pressure – adds further uncertainty, but it is smaller than most fans assume. The biggest obstacle to prediction is not human drama; it is the ball itself.

7 of 15

Recent pre-season model favourites that actually won the league

64%

Title chance of the truly best team in a close race – with perfect knowledge

≈46%

Share of goals involving chance (rebounds, deflections, errors)

Key findings

- 1. Favourites fail half the time.** Across 15 league-seasons with published model probabilities, the pre-season favourite won 7 titles, while the models' own probabilities implied about 9.3 wins.
- 2. Confidence is not certainty.** Favourites rated at 75% or more won 3 of 6 titles in the sample. All three misses came in 2024/25. The sample is small, but it is a warning against reading 80% as "safe".
- 3. Luck sets a hard ceiling.** In my simulation, a team that is genuinely the best by about 7 expected points wins the title only 64% of the time, even when the model knows every team's true strength.
- 4. The human factor costs a further 4–20 points of certainty.** Realistic pre-season uncertainty and in-season shocks reduce the favourite's chance from 64% to between 44% and 60%.
- 5. AI learns fast once the ball rolls.** Opta's model rated Arsenal at 24% in August 2025, 66% by gameweek 11 and 87% by matchday 21. Models are far better at updating than at foreseeing.
- 6. The Bundesliga is the exception that proves the rule.** Bayern's dominance (13 titles in 14 seasons) makes Germany the most predictable top league – until a Leverkusen comes along.

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About the author

Dennis Buchbauer is a betting and predictive analytics specialist focused on sports betting markets, forecasting models and bettor behaviour in the German-speaking market. His previous research includes *The Margin Paradox* (2026), *Predictive Analytics in Sports Betting* and *Sports Analytics in German Football: Can Success Be Predicted?*. More at www.dennis-buchbauer.de.

SECTION 1

1. Introduction: the August question

Few questions in sport generate as much data, debate and money as “who will win the league?” Clubs plan budgets around it, broadcasters build narratives on it, and bettors stake billions on outright markets. Over the past decade, the answer has increasingly been outsourced to machines. Opta’s “supercomputer”, FiveThirtyEight’s SPI model, bank research teams and, most recently, large language models publish title probabilities that are quoted across the media as if they were forecasts of fact.

My earlier paper *Sports Analytics in German Football* asked whether success can be predicted at all. This paper narrows the question to the hardest and most visible prediction in football – the champion – and asks what artificial intelligence has changed. Three questions guide the analysis:

- **Track record:** How accurate have pre-season AI predictions of champions actually been?
- **Limits:** How much of a title race is predictable in principle, and how much is luck?
- **Human factor:** Which human influences – injuries, coaches, crowds, pressure, money – break the models, and how large are they compared with pure chance?

Key terms

Table 1. Definitions.

Term	Meaning in this paper
Title probability	The share of simulated seasons in which a team finishes first. “Liverpool 28.5%” means Liverpool won 2,850 of 10,000 simulated seasons.
Supercomputer	Media term for a Monte Carlo season simulation. Opta’s version combines betting-market odds and its own Power Rankings and plays out the remaining fixtures 10,000 times [1].
Favourite	The team with the highest pre-season title probability.
Calibration	Whether events given X% probability happen about X% of the time. A model can be well calibrated and still “wrong” about the champion most years.
xG (expected goals)	The probability that a shot becomes a goal, based on its characteristics. A measure of chance quality rather than outcomes.

SECTION 2

2. How the machines predict a champion

Every title model rests on the same idea: estimate how strong each team is, translate strength into probabilities for every remaining match, and simulate the season thousands of times. The approaches differ in how they estimate strength. Classical statistical models treat goals as Poisson-distributed events driven by attack and defence ratings [22]. Machine-learning models add hundreds of features – ratings, player data, xG – and learn non-linear relationships; the winner of the 2017 Soccer Prediction Challenge used gradient-boosted trees trained on more than 200,000 matches [23]. Commercial models such as Opta’s blend betting-market odds with proprietary ratings [1], which means their numbers are not fully independent of the bookmakers.

Table 2. Families of title-prediction models.

Approach	How strength is estimated	Typical strength	Typical weakness
Poisson / Elo models	Goals and results, time-weighted	Transparent, robust	Slow to react to squad change
Machine learning	Ratings, xG, player and tracking data	Captures non-linear effects	Needs data; can overfit
Market-based (e.g. Opta)	Betting odds plus power ratings	Absorbs crowd wisdom	Inherits market biases
Large language models	Text knowledge plus reasoning	Context, narratives	Herding; no edge over markets

The newest entrants are large language models. A study of four frontier LLM agents forecasting all 104 matches of the 2026 World Cup found that none beat the betting market’s Brier score, that the agents picked the same favourite in 92% of matches, and that betting against the market lost money for every agent [12]. Language models can explain football; they do not yet predict it better than prices do.

SECTION 3

3. The scorecard: predictions versus champions

Figure 1 collects every pre-season title probability I could verify for Europe’s top leagues since 2019/20 – 15 league-seasons from Opta, Stats Perform and FiveThirtyEight [2, 3, 4, 5, 6]. The pre-season favourite won **7 of 15** titles. Summing the models’ own probabilities, they should have won about **9.3**.

The supercomputer's favourite won 7 of 15 recent titles

Pre-season title probability (Opta / Stats Perform / FiveThirtyEight) vs. actual champion

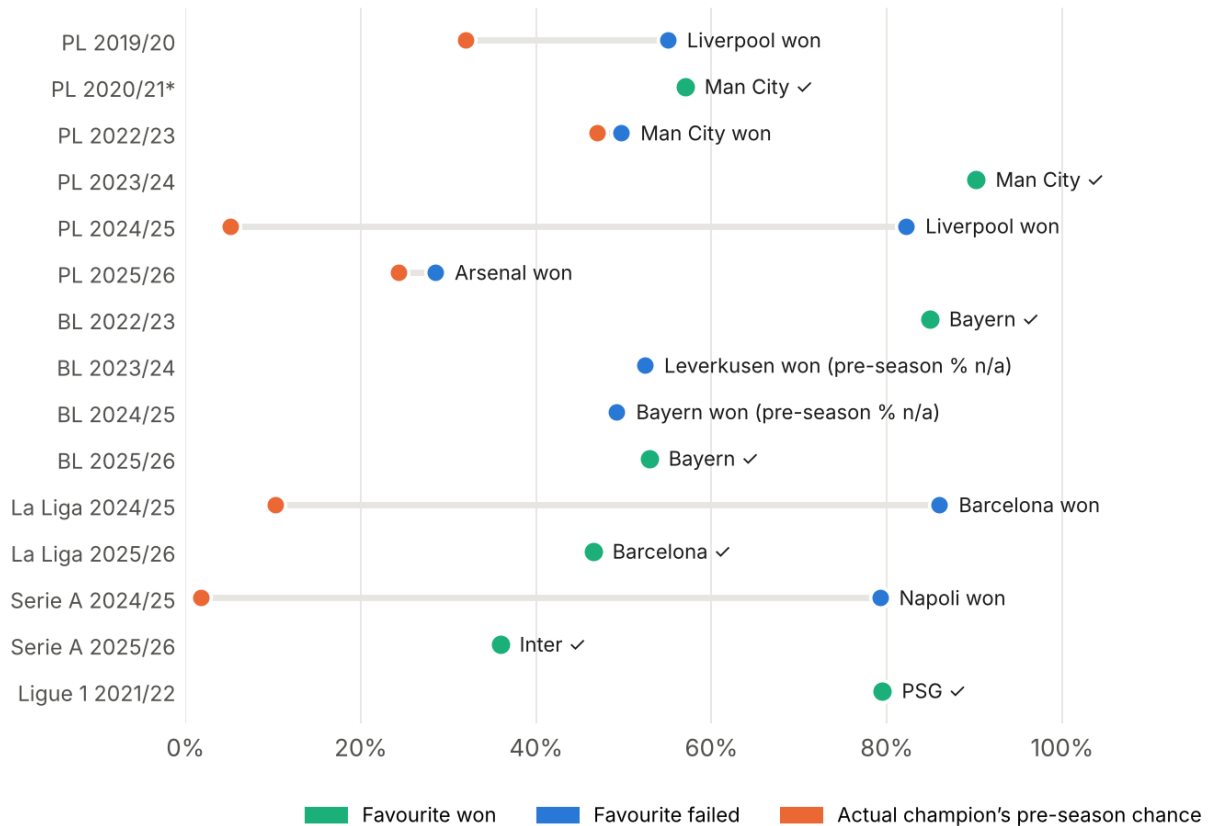


Figure 1. Pre-season title probability of the favourite and of the eventual champion. * PL 2020/21: FiveThirtyEight after matchday 1. Serie A 2024/25: Napoli shown at the reported upper bound of 1.8%. Bundesliga 2023/24 and 2024/25: champion's pre-season probability not published in the sources found. Sources: [1–6].

Two patterns stand out. First, the favourite's failure rate is not concentrated in close races. When a model was at least 75% confident, the favourite won only 3 of 6 titles – Manchester City 2023/24, Bayern 2022/23 and PSG 2021/22 – while City, Real Madrid and Inter all failed in 2024/25. Six cases are far too few to prove that the models are over-confident, and a single unusual season drives the result. But they show that a headline figure of 80% leaves plenty of room for surprises.

Second, the models systematically underrated a rising team. Opta gave Arsenal 0.13% in 2022/23, 4.1% in 2023/24 and 12.2% in 2024/25; Arsenal finished second each time before winning the title in 2025/26 with a pre-season rating of 24.3% [2]. Models anchored on past results are, by design, slow to recognise genuine improvement.

Arsenal: the model learned slowly, the team improved quickly

Opta pre-season Premier League title probability and final position

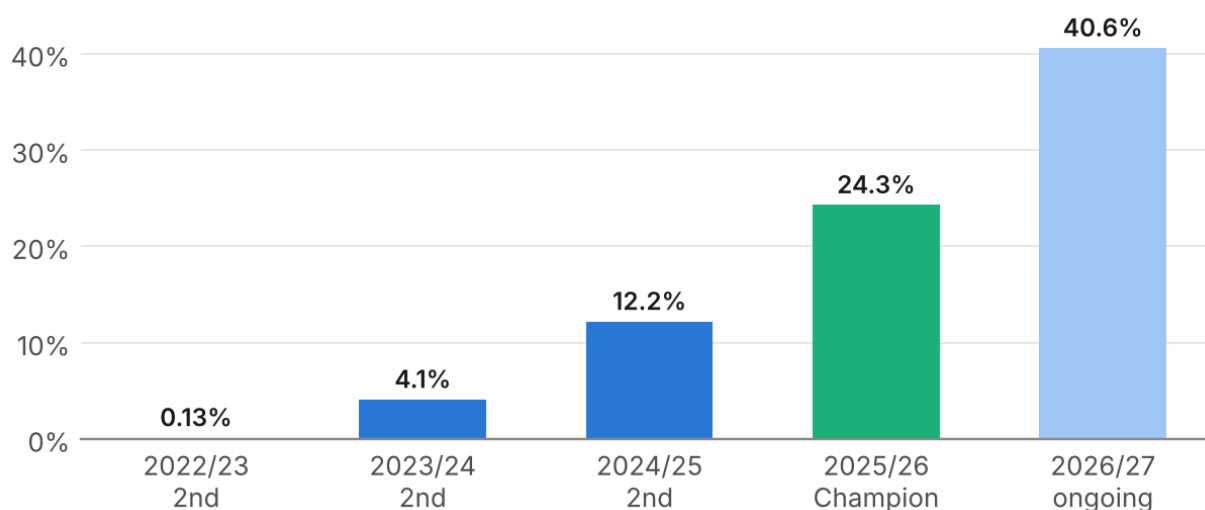


Figure 2. Arsenal's Opta pre-season Premier League title probability and final league position, 2022/23–2026/27. Source: The Analyst / Opta [2].

Table 3. Notable title surprises against model and market expectations. Sources: [2–5], [8], [38].

Season	League	Model favourite	Champion	Champion's pre-season chance
2015/16	Premier League	Chelsea (bookmakers 7/4)	Leicester City	5000/1 at bookmakers; ≈0.004% in Oxera's model
2023/24	Bundesliga	Bayern 52.4%	Bayer Leverkusen (unbeaten)	Not published; Stuttgart (2nd) won 0.01% of simulations
2024/25	Premier League	Man City 82.2%	Liverpool	5.1%
2024/25	La Liga	Real Madrid 86%	Barcelona	10.3%
2024/25	Serie A	Inter 79.2%	Napoli	≤ 1.8%
2025/26	Premier League	Liverpool 28.5%	Arsenal	24.3%

Tournaments: even harder

Knock-out tournaments are less predictable still. No pre-tournament favourite in the last five major tournaments was rated above about 21%, and the consensus favourite failed four times in a row – Brazil in 2018, France at Euro 2020, Brazil in 2022 and England at Euro 2024 [11, 10]. The 2026 World Cup broke the pattern: Opta (16.1%), Goldman Sachs (26%), the Innsbruck machine-learning ensemble (14.5%) and the betting market all made Spain favourite, and Spain beat Argentina 1–0 after extra time [11]. Even then, the models gave the eventual winner well under a one-in-three chance.

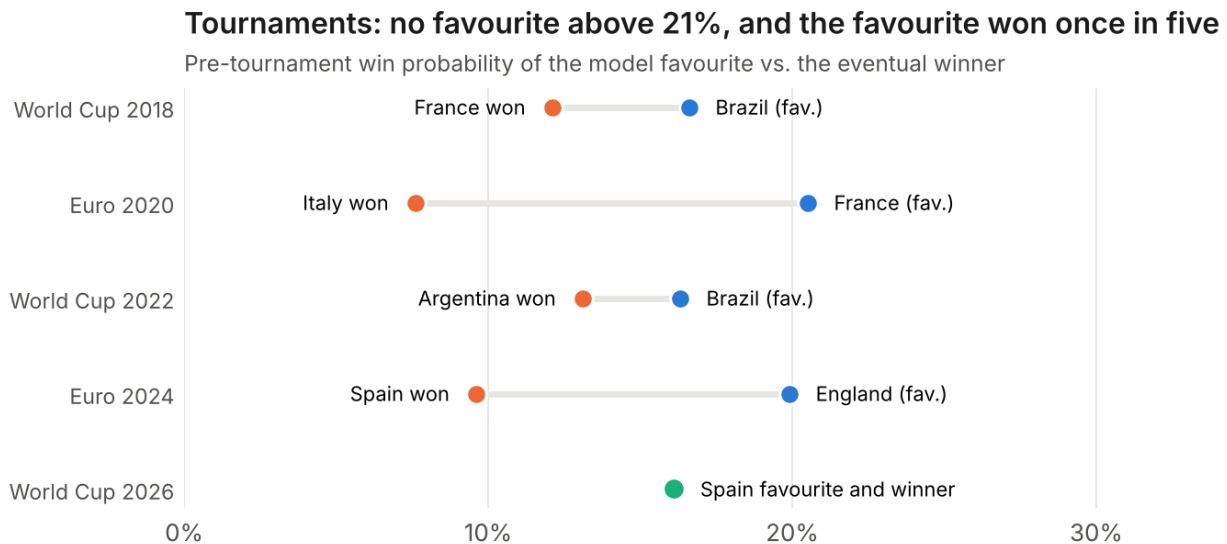


Figure 3. Pre-tournament probability of the model favourite and of the eventual winner. Sources: Zeileis et al. bookmaker consensus (2018), Opta (2020–2026) [10, 11].

The honest benchmark for a forecast is not whether the favourite wins but whether probabilities are realistic. On that measure the best models are useful: for the 2018 World Cup, the bookmaker consensus ranked teams with a Spearman correlation of 0.70 against the final outcome, ahead of Elo (0.59) and the FIFA ranking (0.41) [10]. Predictions of whole league tables are weaker: in a 2014/15 comparison of 60 Premier League table predictions, only two of four betting lines beat the naive rule “same order as last season” [9].

SECTION 4

4. Why football resists prediction

Football is structurally harder to predict than other major team sports. Ben-Naim, Vazquez and Redner measured the share of games won by the weaker team across a century of data: 45.2% in English football, against 36.5% in the NBA and 36.4% in the NFL [13]. A value of 50% would be a pure coin flip. Aoki, Assunção and Vaz de Melo, analysing 1,503 seasons in four sports, likewise find that basketball has the strongest skill component, while football sits much closer to a random league [14].

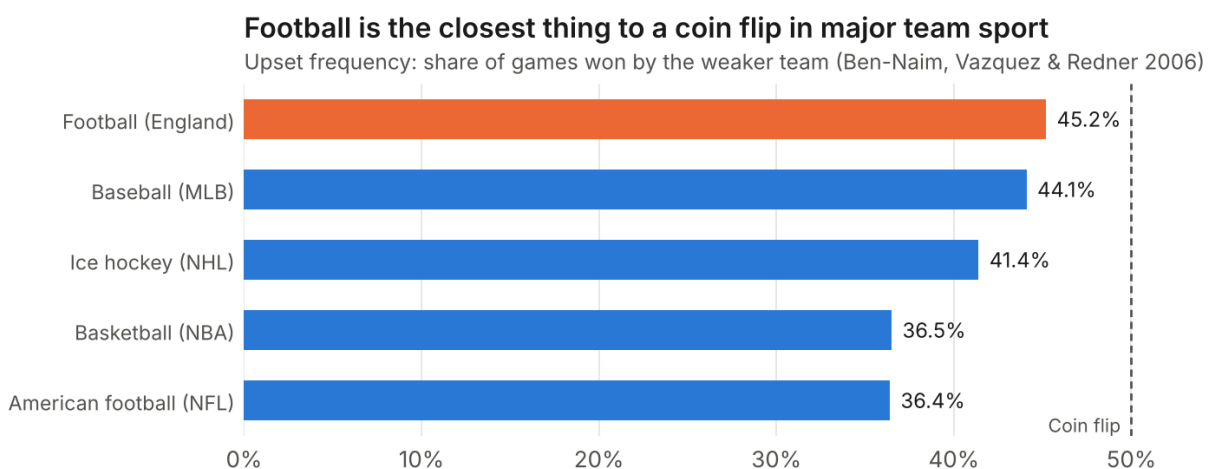


Figure 4. Upset frequency by sport: share of games won by the team with the worse record. Source: Ben-Naim, Vazquez & Redner (2006) [13].

The reason is the scoring structure. Goals are rare, so single events decide matches. Wunderlich and Memmert coded 7,263 Premier League goals and found that about 46% involved randomness – rebounds, deflections, long-range shots, defensive errors – and that more than 60% of matches end as a draw or with a one-goal margin [15]. Lames reports chance involvement of 46.9% for Bundesliga goals [16]. When nearly half the decisive events contain luck, no model can reach certainty about the outcome of a single match – and a title is the sum of 34 or 38 of them.

How much does that luck matter over a season? Holzmeister and Johannesson decompose over- and under-performance relative to expected points in seven top leagues: about 40% is skill and 60% luck, which corresponds to a standard deviation of roughly 7 points from luck over a 38-match season and an average displacement of 1.8 table positions [17]. David Spiegelhalter estimated that Manchester United, the 2006/07 champions, had only a 53% probability of genuinely being the best team that season [18]. Heuer and Rubner show that team strength is essentially constant within a season, yet goal difference in the first half of a season correlates with the second half at only about 0.53–0.55 [19, 20].

SECTION 5

5. Original analysis: how much of a title race is predictable?

To separate predictable from unpredictable uncertainty, I built a Monte Carlo simulator of an 18-team league in Bundesliga format (34 matchdays, double round-robin). Each team has a single true strength; goals follow Poisson distributions with a home advantage. Every scenario is simulated 20,000 times. The simulator lets me do what real data cannot: give a model *perfect* knowledge of every team's true strength and measure what is left.

Model calibration

Average 2.98 goals per match, 43% home wins and 21% draws – close to typical Bundesliga values. Expected points range from 72 for the strongest to 24 for the weakest team.

Within-season luck produces a standard deviation of 7.0 points per team – almost exactly the ~7 points from luck estimated empirically by Holzmeister & Johannesson [17]. The simulated league is therefore realistically noisy, not artificially random.

Result 1: the luck ceiling

Figure 5 shows how often the genuinely best team wins the title, depending on how much better it is than the second-best team. With a lead of about 3 expected points – a typical two-horse race – the best team wins only 43% of seasons. With about 7 points it wins 64%. Even a dominant side 16 points better than its nearest rival fails in 6% of seasons. This is the ceiling for *any* predictive model, however sophisticated: it applies with perfect information.

Even a perfect model cannot name the champion

Simulated probability that the truly best team wins the title, 20,000 seasons per point

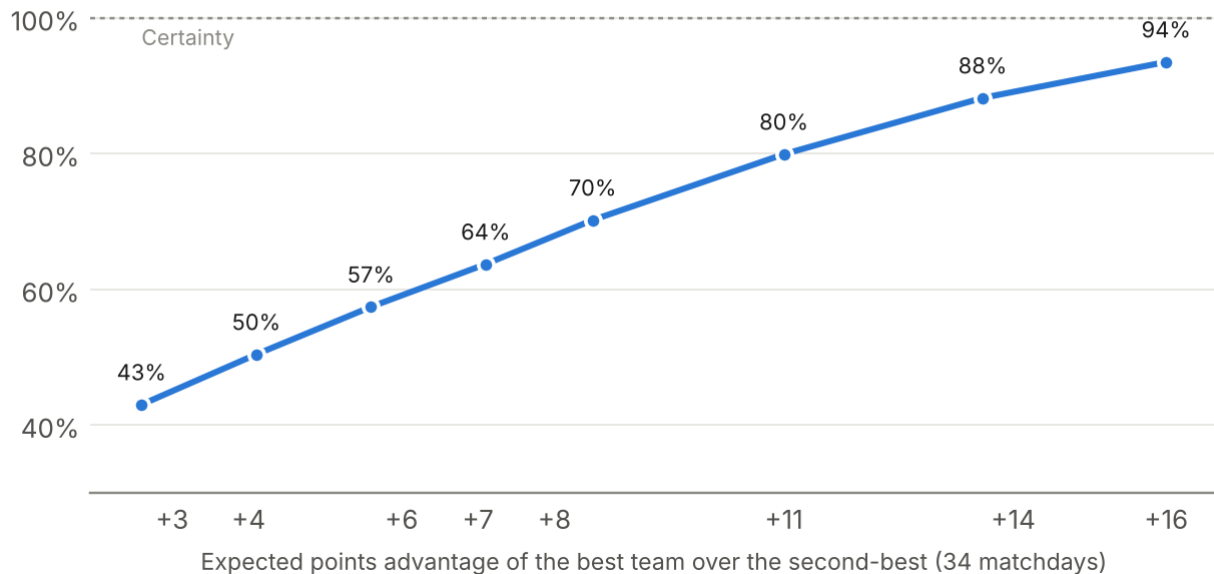


Figure 5. Probability that the truly best team wins the title, as a function of its expected points advantage over the second-best team. Author's simulation, 20,000 seasons per point.

Result 2: the predictability budget

Real models do not know true strength; they estimate it, and in-season events change it. I therefore added two layers to a two-horse race with a true lead of about 7 points. The first is **pre-season estimation error** – the uncertainty created by transfers, new coaches and tactical change. The second is **in-season shocks** – injury crises, sackings, form collapses or surges that permanently shift a team's strength during the season.

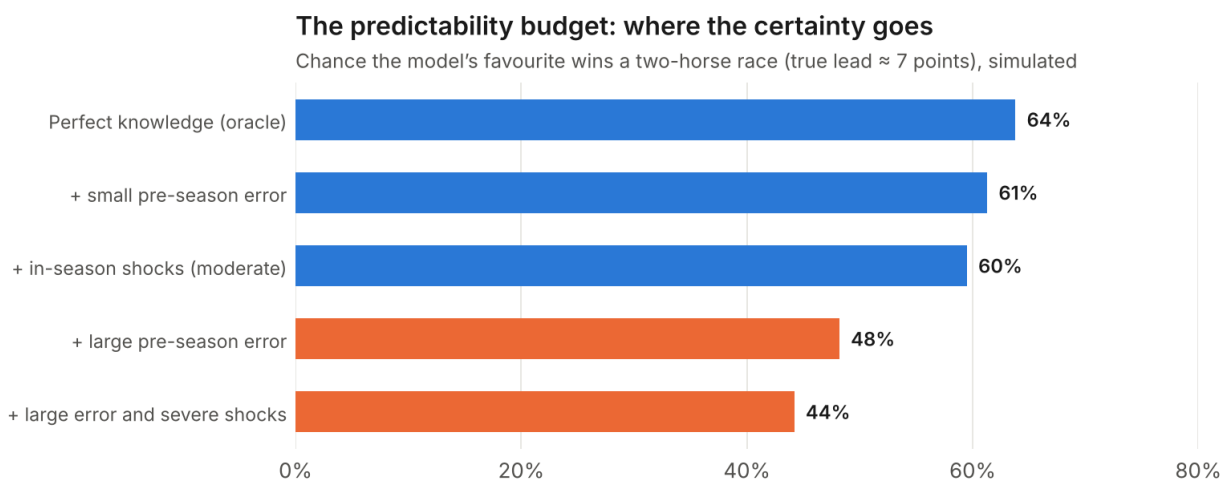
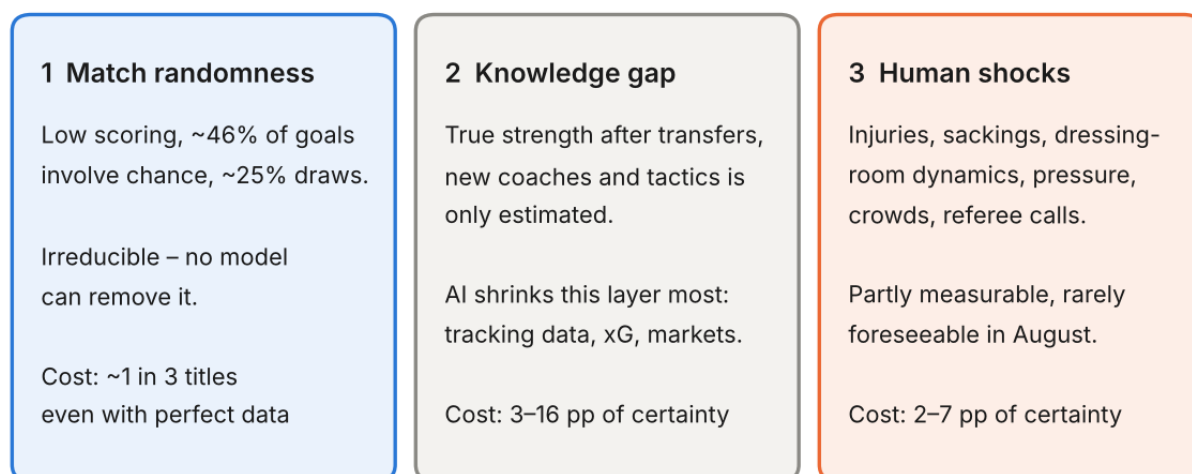


Figure 6. Chance that the model's pre-season favourite wins the title. Small/large estimation error: standard deviation 0.10/0.20 in log-goal units; moderate/severe shocks: 3%/5% chance per team per matchday of a permanent strength change of SD 0.12/0.20. Author's simulation.

The result reframes the human-factor debate. Match randomness alone costs the favourite about 36 percentage points of certainty. Realistic in-season shocks cost a further 2–7 points. The largest controllable layer is the pre-season knowledge gap: a large estimation error lowers the favourite's chance to 48%. This is

exactly where AI adds value – better player data, tracking data, xG and market information shrink the knowledge gap – but it cannot touch the first and largest layer.

Three layers of uncertainty between a model and the trophy



Costs: percentage points of title certainty lost, from the author's simulation (Section 5)

Figure 7. Three layers of uncertainty between a model and the trophy. Author's framework, quantified with the simulation in this section.

Result 3: when does the table tell the truth?

Finally, I measured how quickly the league table converges to true strength. After 9 matchdays, the leader becomes champion in only 44% of simulated seasons; at the halfway point (matchday 17) in 54%; and with nine games left (matchday 25) in 68%. Even the final table only correlates at 0.90 with true strength. This mirrors Bond et al., who show that partial standings in English football converge to the final table along a predictable curve [21].

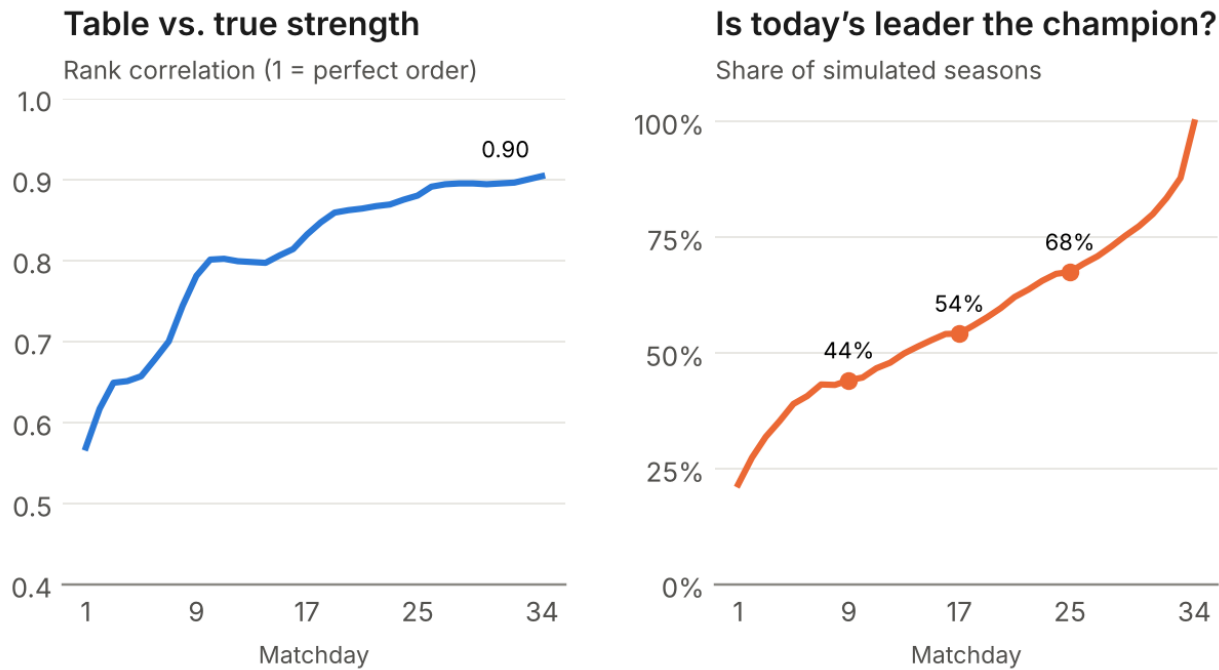


Figure 8. Left: rank correlation between the table and true team strength by matchday. Right: share of seasons in which the current leader becomes champion. Author's simulation.

SECTION 6

6. The human factor: real, measurable, but not dominant

If luck is the largest source of uncertainty, the human factor is the most visible. The evidence suggests that it matters – but in more predictable ways than its reputation suggests.

Coaches

Sacking the coach is football's favourite intervention, yet the evidence for its effect is thin. Heuer et al. analysed Bundesliga dismissals from 1963 to 2009 and found “basically no effect” on subsequent performance [24]. Van Ours and Van Tuijl showed for the Netherlands that teams improve after a change – but a control group of clubs that nearly sacked their coach improved just as much [25]. Clubs typically sack at around 1.0 point per game, below their normal level of about 1.3, so the famous “new manager bounce” is largely regression to the mean [26] – something a statistical model anticipates automatically.

Injuries

Injuries are the most quantifiable human factor. In the Premier League, about 136 injury days cost one league point and about 271 days one league place; injuries explain about 21% of the gap between where a club “should” finish based on squad value and where it does [27]. UEFA's long-running injury study links lower injury burden and higher player availability with higher league ranking [28]. Leicester's 2015/16 miracle coincided with the fewest injuries in the league and the fewest players used [38]. Injuries are partly forecastable from workload and age data, but their timing is not.

Crowds and pressure

The COVID-19 ghost games produced a natural experiment no model had been trained for. Before the shutdown, Bundesliga home teams won 43% of matches; in the first 56 games without spectators, only 21% [29]. In the Premier League 2020/21, home teams won a record-low 37.9% and away wins exceeded home wins for the first time [30]. Pressure also shows up in individual actions: in penalty shoot-outs, success falls from 87% on the first kick to 73% on the fourth [31]. Title races have their own version – Arsenal led the 2022/23 Premier League by eight points with nine games left and then dropped 15 points in eight matches.

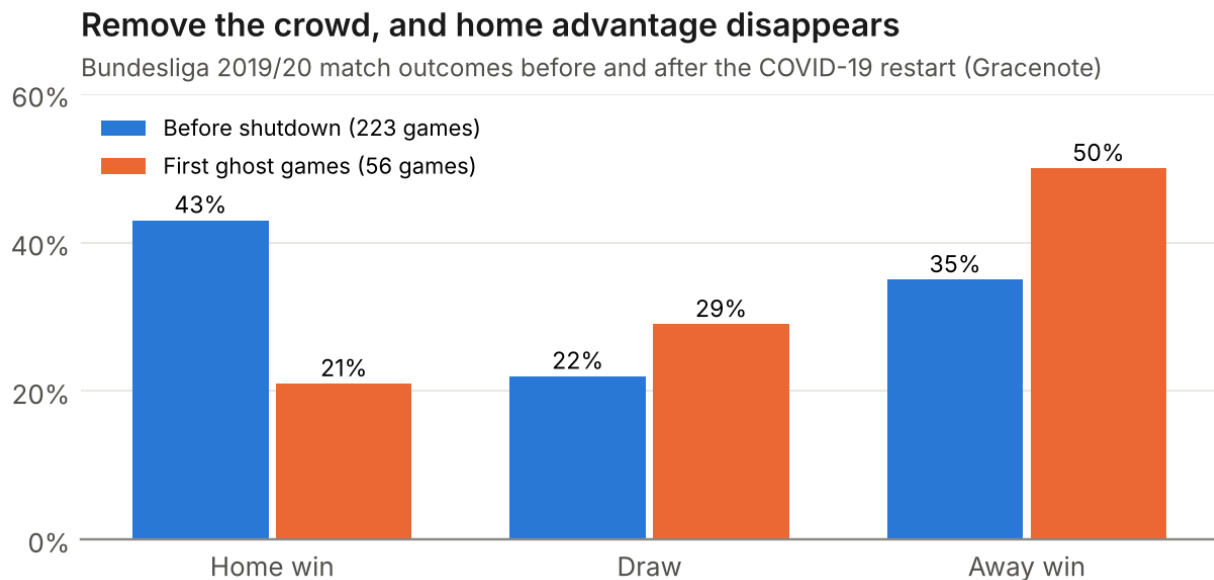


Figure 9. Bundesliga 2019/20: share of home wins, draws and away wins before the shutdown and in the first ghost games. Source: Gracenote via Reuters [29].

Money: the slow, strong signal

Over the long run, money is the most powerful predictor of all. Kuper and Szymanski show that relative wage spending explains around 90% of the variation in English clubs' average league positions over a decade [32]. Within a single season the relationship is much weaker – one Premier League season shows an R^2 of roughly 0.5 – which leaves room for the human factor and luck. Liverpool illustrates both sides: champions with 84 points under a new head coach in 2024/25, fifth with 60 points a year later despite heavy summer spending [37].

Table 4. The human factor: evidence and predictability.

Human factor	Evidence	Predictable by AI?
Coach changes	No measurable causal effect; the 'bounce' is regression to the mean [24, 25]	Largely yes
Injuries	136 injury days $\approx$ 1 point; explain $\sim$ 21% of under-/over-performance [27]	Risk yes, timing no
Crowds	Home wins 43% $\rightarrow$ 21% in Bundesliga ghost games [29]	Once observed, yes
Pressure	Penalty success 87% $\rightarrow$ 73% from 1st to 4th kick [31]	Hardly
Money	Wages explain $\sim$ 90% of positions over a decade, far less per season [32]	Yes, long-term
Over-performance	Leverkusen 2023/24: 89 goals from 75.0 xG; 17 goals after the 89th minute [37]	No

SECTION 7

7. Where AI genuinely helps

The verdict on pre-season prediction should not hide where AI performs well. Its real strength is **updating**. Opta's model rated Arsenal at 24% in August 2025, 44% after gameweek 7, 66% after gameweek 11 and 87% at matchday 21 [7]. In the 2023/24 Bundesliga it moved Bayern from 52% to 80% in November, then switched its favourite to Leverkusen in January – three months before Leverkusen clinched the title [3].

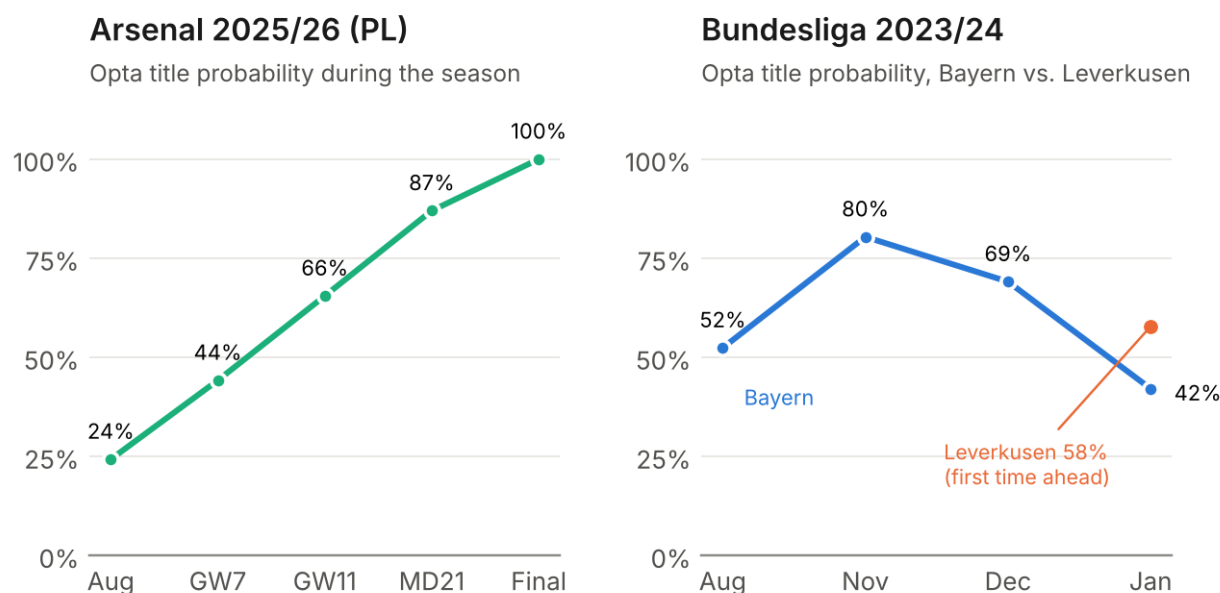


Figure 10. Opta title probabilities during two seasons: Arsenal 2025/26 and the Bundesliga 2023/24. Sources: Sports Illustrated / Opta [7], The Analyst / Opta [3].

Better inputs are the second strength. Expected-goals ratios predict future points better than goals or points themselves [36], and the Bundesliga's AWS-powered xGoals model is trained on about 40,000 historical shots [35]. Inside clubs, AI is moving from prediction to decision support: Google DeepMind's TacticAI, developed with Liverpool, was trained on 7,176 Premier League corner kicks, and club experts preferred its tactical suggestions over existing set-ups in 90% of blind comparisons [34]. Brentford, Brighton and FC Midtjylland have built entire club strategies around statistical modelling.

These uses share a pattern: AI is strongest where it informs many small, repeated decisions – corners, transfers, rotation, in-play probabilities – and weakest at the single, high-variance prediction the public cares most about.

SECTION 8

8. The German perspective: Bundesliga, the predictable exception

The Bundesliga is the most predictable of Europe's top leagues – for structural rather than technological reasons. Bayern Munich won 11 consecutive titles from 2012/13 to 2022/23 and, after Leverkusen's unbeaten 2023/24 season, two more; that is 13 titles in 14 seasons [5]. Research from ZEW found that competitive balance in the Bundesliga fell by about 30% between 1998 and 2016 [33]. In such a league, “Bayern” is a well-calibrated forecast most years, with or without AI. For 2026/27, Opta rates Bayern at 76.7% [3].

Yet the Bundesliga also produced the clearest modern case of a model being overtaken by reality. In August 2023, Opta gave Bayern 52.4%; VfB Stuttgart, who eventually finished second ahead of Bayern, won the title in just 1 of 10,000 simulations [3]. Leverkusen scored 89 goals from 75.0 expected goals, conceded 24 and scored 17 goals after the 89th minute across competitions [37] – a combination of quality, coaching and fortune that no pre-season model could have assembled. One season later, the model over-corrected and made Leverkusen favourite (49.2%); Bayern won by 13 points.

For German bettors, the implication is practical. Outright markets on the Bundesliga winner are often priced efficiently because the favourite is so clear; the uncertainty – and the opportunity for informed judgement – lies in second place, the European places and relegation, where the gap between teams is smaller and the human factor matters more.

SECTION 9

9. Verdict and outlook

So, is it possible today to determine the league champion with the power of AI? **No – and it will not become possible with better technology alone.** AI can tell us who is most likely to win, and it does so more accurately than pundits, fans and simple rules. But the outcome of a football season contains a large, irreducible element of chance. My simulation shows that even a model with perfect knowledge would name the wrong champion in roughly one of three close races; real models, facing transfers, injuries and coaching changes, miss more often. The track record – 7 of 15 recent favourites – fits that picture.

Is the human factor too strong? It is real, and in individual seasons it decides titles. But it is not the main reason models fail. Coaching changes mostly reflect regression to the mean; injuries are partly measurable; crowd effects, once observed, are modelled quickly. The deepest limit is the nature of the game: a low-scoring sport in which nearly half the goals involve chance.

- **Read probabilities, not predictions.** “Arsenal 40.6%” for 2026/27 means Arsenal are more likely *not* to win the title than to win it.
- **Judge models by calibration.** A good model's 60% favourites should win about six times in ten – no more.
- **Trust the updates more than the August numbers.** By mid-season, models absorb information much faster than human intuition does.
- **Expect AI to win small decisions, not big headlines.** Set pieces, recruitment and in-play pricing are where the technology already changes football.

FAQ

10. Frequently asked questions

Can AI predict who will win the Premier League or Bundesliga?

AI can estimate each team's probability of winning, but it cannot determine the champion. In 15 recent league-seasons, the pre-season model favourite won 7 titles.

How accurate is the Opta supercomputer?

Its favourites have won about half of recent titles, slightly fewer than its own probabilities implied. It is most useful during the season, when it updates quickly – for example from 24% to 87% for Arsenal in 2025/26.

How much luck is involved in winning a league?

Around 46% of goals involve an element of chance, and roughly 7 points of a team's season total can be attributed to luck. In a close title race, even the best team wins only about 50–65% of the time.

Is the human factor still too strong for AI?

The human factor – injuries, coaching changes, crowds and pressure – reduces predictability, but chance in the game itself is the bigger obstacle. Many human effects, such as the new-manager bounce, are statistically predictable.

Did AI predict the 2026 World Cup winner?

Yes – Opta, Goldman Sachs, the Innsbruck ML model and bookmakers all made Spain favourite, and Spain won. But Spain's pre-tournament probability was only 14–26%, and favourites had failed in the four previous tournaments.

Is the Bundesliga easier to predict than other leagues?

Yes. Bayern's dominance – 13 titles in 14 seasons – makes the Bundesliga the most predictable top league, although Leverkusen's unbeaten 2023/24 season showed that even long-term certainty can break.

METHODOLOGY

11. Methodology and limitations

The scorecard in Section 3 uses pre-season title probabilities published by Opta/Stats Perform and FiveThirtyEight between 2019 and 2026, as reported by the publishers or by media outlets syndicating their figures. Seasons for which no verifiable pre-season probability was found are excluded; one figure (PL 2020/21) was published after the first matchday. With 15 observations, the hit rate and the calibration comparison are indicative, not statistically conclusive.

The simulation in Section 5 is a stylised model: one strength parameter per team, independent Poisson goals, constant home advantage (0.12 in log-goal units) and a baseline of 1.40 goals per team per match. Team strengths are evenly spaced (± 0.75); title-race scenarios add a strength bonus to the best team. Estimation error and in-season shocks are modelled as normally distributed perturbations. The model is calibrated to realistic goal, result and luck levels but does not represent any specific season. Code and parameters are available from the author.

Several figures in Sections 4 and 6 come from secondary summaries of the underlying studies, as noted in the references. Readers relying on individual numbers should consult the primary sources.

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